Supplementary Materials for

Effects of Velocity on Evaporation Rate of Nanoscale Water Droplets

Elisa Y.M. Ang[a,*], *Peng Cheng Wang*[a], *William Toh*[b], *Teng Yong Ng*[b]

a. Engineering Cluster, Singapore Institute of Technology, 10 Dover Drive, Singapore 138683 Singapore

b. School of Mechanical and Aerospace Engineering, Nanyang Technological University, 50 Nanyang Avenue, Singapore, 639798, Singapore

Correspondence to: elisa.ang@singaporetech.edu.sg

This supplementary document includes:

Section S1: Simulation run on evaporation dynamics under NVE conditions

Section S2: Repeat runs to demonstrate consistency of results

Supplementary Video SV1: The droplets dynamics at 0m/s, 100m/s and 1200m/s.

Section S1: Simulation run on evaporation dynamics under NVE conditions

In the paper, the canonical (NVT) ensemble is assumed, where the system is assumed to be under constant temperature and volume condition. The rationale for isothermal assumption is the assumed scenario of having a small nanodroplet surrounded by a large body of air, which therefore acts as the thermal reservoir. The small nanodroplet is assumed to be the body moving through air, and therefore any heating near the nanodroplet will be dissipated quickly with the large body of air surrounding the nanodroplet. However, due to computational limits, a finite body of air needs to be assumed in this case, therefore, to model the large body of air, an NVT condition is enforced. In addition, due to computational limits, instead of moving the nanodroplet, the air molecules are moved instead. Due to the small volume of the nanodroplet, heat transfer via convection and conduction can occur very fast through the droplet, which therefore also justify the isothermal assumption through the entire simulated system. In this assumption, the air temperature remains constant, but the pressure around the droplet increases, as demonstrated in **Figure 7 of the Manuscript**. This simulated case is largely relevant for cases where the droplet is ejected at high speed, in which the relative velocity of the surrounding air molecules with respect to the droplet is high, but the ambient temperature remains generally unchanged. This is the case under consideration in this simulation.

If one would like to consider a rise in temperature of ambient air due to heating up of the air from the high velocity, one can no longer assume the canonical ensemble. In this case, the evaporation mechanics should be considered using the micro-canonical ensemble (NVE) condition. This is the method used by Hou et al (Hou et al., 2023), who studied the evaporation of nitrate-containing

nanodroplets within a body of hot ambient gases. To look at the theoretical increase in temperature considering a large volume of droplets travelling at high speed, such that the reservoir of air is all heated up, or for a stationary droplet within high speed ambient gas, a separate simulation with NVE condition is ran at the highest speed, 1180 m/s occurring when the applied force is 1e-3 kcal/mol-Å (refer to **Manuscript Figure 2**). The result for this separate run is presented in Figure S1.1 below. One can see that in this case, the air molecules heat up very fast. This causes evaporation/flashing behaviour of the surface water molecules, which therefore results in the droplet evaporating to less than 30% of its volume in 5ns. However, this case is only valid if the atmospheric air is heated up. If a single droplet is travelling through a large ambient reservoir of air at high speed, this is unlikely to be the case. In this case, the canonical ensemble, as presented in this paper, is a better approximation.

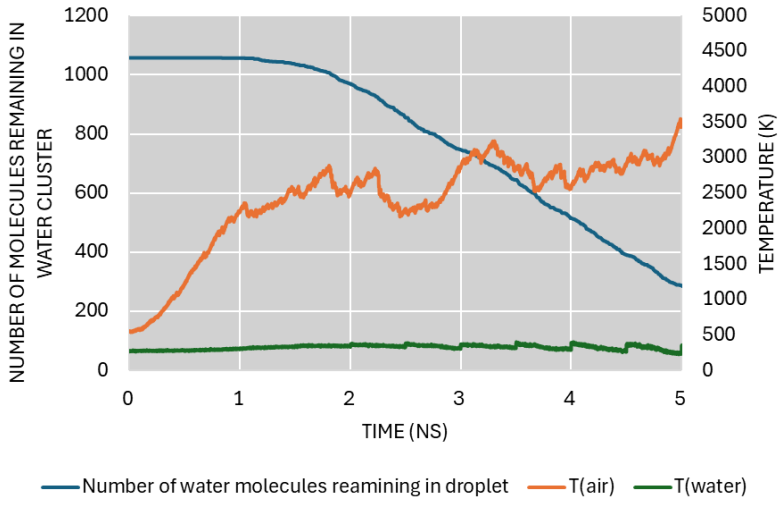


Figure S1.1 Results of simulation run with NVE condition, which assumes droplet is within high speed ambient gas, or a large volume of droplets travelling at high speed

Section S2: Repeat runs to demonstrate consistency of results

As described in the main manuscript, the measurement of evaporation rate in this contribution is obtained via monitoring the change in number of remaining water molecules over time. Each simulation is carried out once, however, a check was done to ensure the consistency of data by repeating the simulation thrice for the case when force = 0 and force = 1e-3 kcal/mol-Å, for different initial configurations. Results of the repeat runs are presented in Figure S1.1. As can be seen from Figure S1.1, the results are very repeatable.

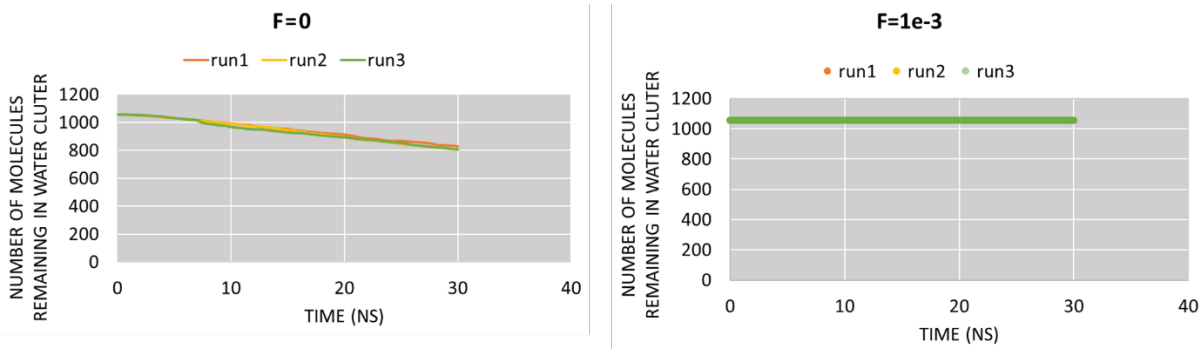


Figure S2.1 Repeat runs to demonstrate consistency of results for F=0 and F=1e-3

References

Hou, D., Wang, G., Gao, J., & Luo, K. H. (2023). Molecular dynamics study on evaporation of metal nitrate-containing nanodroplets in flame spray pyrolysis. *Nanoscale*, *15*(12), 5877–5890. https://doi.org/10.1039/D3NR00060E